

Basic Principles of Medicine 2

Module: Digestion and Metabolism

Lecture: DM 31

Alcohol and Xenobiotic Metabolism in Liver

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Objectives: Alcohol and Xenobiotic Metabolism in Liver

SOM. MK.I.BPM2.3.DM.2.BCHM.1288	Describe role of cytochrome P450 enzymes in drug metabolism and specify roles CYP 2E1 and CYP 3A4. Discuss physiological relevance of induction of CYP 2E1 by ethanol.
SOM. MK.I.BPM2.3.DM.2.BCHM.1289	Describe mechanism of acetaminophen toxicity and the molecular basis for use of N-acetyl cysteine for management.
SOM. MK.I.BPM2.3.DM.2.BCHM.1290	Describe general features of ethanol and describe routes of hepatic ethanol metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1291	Discuss role of alcohol dehydrogenase, acetaldehyde dehydrogenases and microsomal ethanol oxidizing system (MEOS) in hepatic ethanol metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1292	Explain effects of high ethanol consumption on liver gluconeogenesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1293	Describe metabolic basis for acute and chronic toxic effects of ethanol abuse highlighting the role of acetaldehyde.
SOM. MK.I.BPM2.3.DM.2.BCHM.1294	Review the significance of drugs that inhibit aldehyde dehydrogenase (disulfiram) and alcohol dehydrogenase (fomepizole)
SOM. MK.I.BPM2.3.DM.2.BCHM.1295	Explain metabolic basis of fatty liver.
SOM. MK.I.BPM2.3.DM.2.BCHM.1296	Differentiate between methanol or ethylene glycol poisoning using laboratory findings (toxic compounds and anion gap) and clinical features. Explain the metabolic basis for management of these disorders.

Recommended Reading

- Lippincott Illustrated Reviews (**Lippincott**) 9th edition; Chapter 23: Section: Alcohol related hypoglycemia;
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>; 23.
 - [Metabolic Effects of Insulin and Glucagon | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Mark's Basic Medical Biochemistry: A clinical approach (**MBMB**) 6th edition; Chapter 33; [Metabolism of Ethanol | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Medical Education | Health Library \(lwwhealthlibrary.com\)](#)
- **Alcoholic fatty liver:** <https://pubmed.ncbi.nlm.nih.gov/22701235/>: Molecular mechanism of alcoholic fatty liver, Karuna Rasineni and Carol A Casey, Indian J Pharmacol. 2012 May-Jun: 44(3): 299-303

Alcohol and Xenobiotic Metabolism in Liver

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Xenobiotics

Xenobiotics are chemicals that are foreign to the body.

Xenobiotics metabolized by cytochrome P450 systems
in intestinal epithelial cells and hepatocytes

Drug metabolism in the liver
makes the molecule **more polar** for **excretion** by kidneys.

Metabolism of drugs in liver occur in two phases:

- **Phase I** involves **cytochrome P450** isozymes.
- **Phase II** makes molecules more polar and **follows** in many cases **phase I**

but **some drugs** are made more **water-soluble** by using only **phase II**

Cytochrome P450 system

CYP reductase and **CYP monooxygenase** work together.

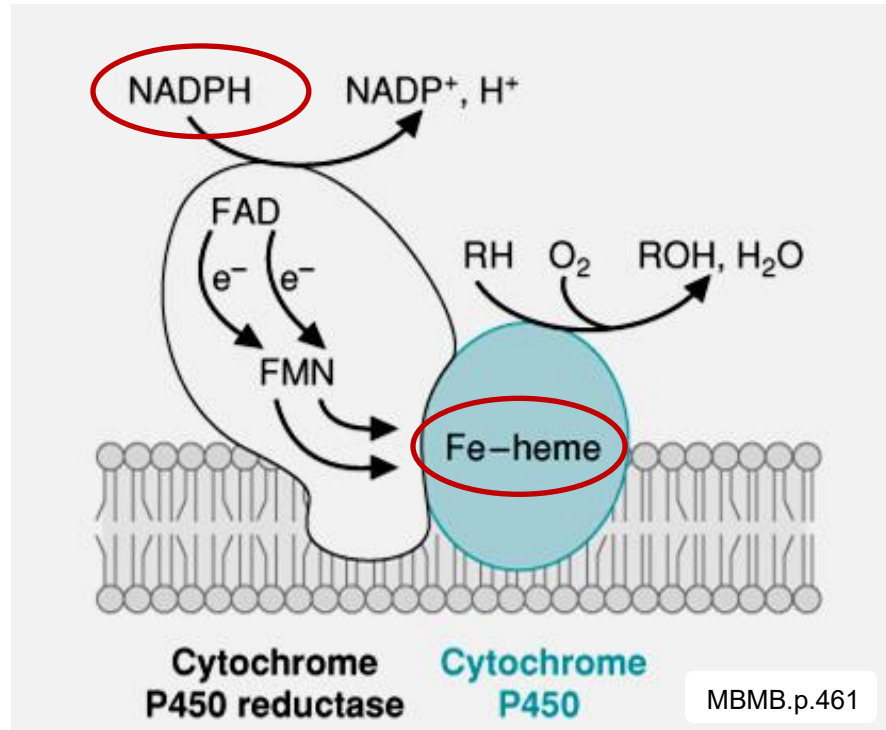
**Cytochrome P450
reductase**

uses **NADPH**

Cytochrome P450

contains **heme**

And often forms hydroxyl group

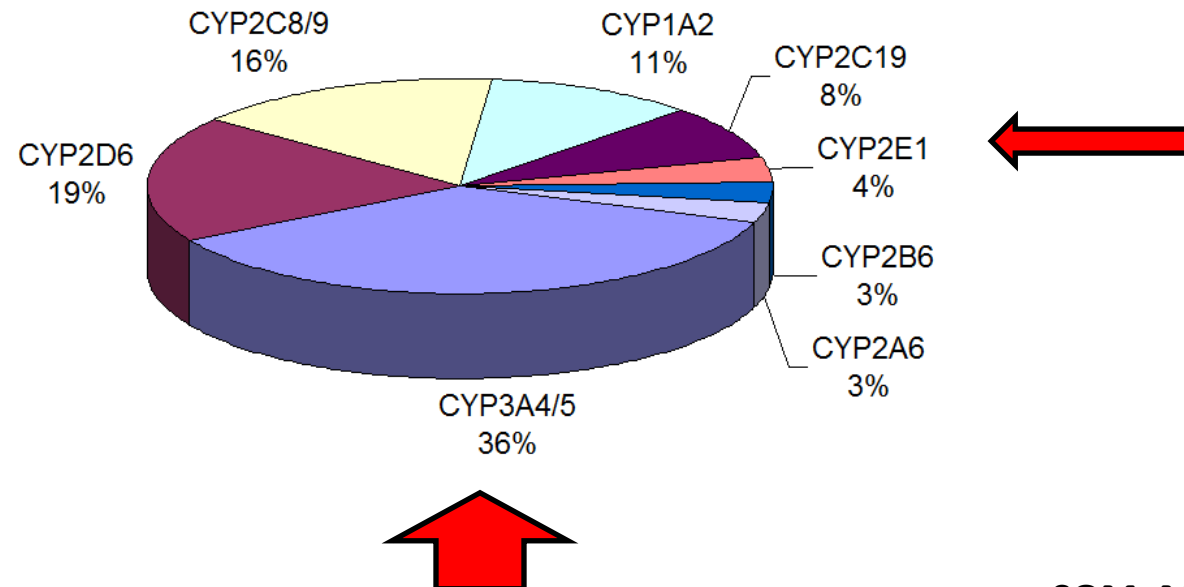


CYP inactivate **drugs**, activate **pro-drugs**, or may convert **drugs** to a **toxic metabolite**. **Free Radicals** are formed.

Cytochrome P450 (CYP) isozymes in hepatocytes

CYP3A4 acts on a variety of therapeutic drugs.

CYP2E1 specific for ethanol metabolism.



The activity of cytochrome P450 (CYP) can be increased or decreased by drugs or natural compounds.

Stimulated synthesis of CYP

Some drugs metabolized by CYP can stimulate synthesis of cytochrome P450 (Inducers)

Ethanol induces CYP 2E1

High CYP 2E1 levels interferes with acetaminophen metabolism

Inactivation of CYP

Grapefruit juice inhibits CYP 3A in **intestinal epithelial cells.**



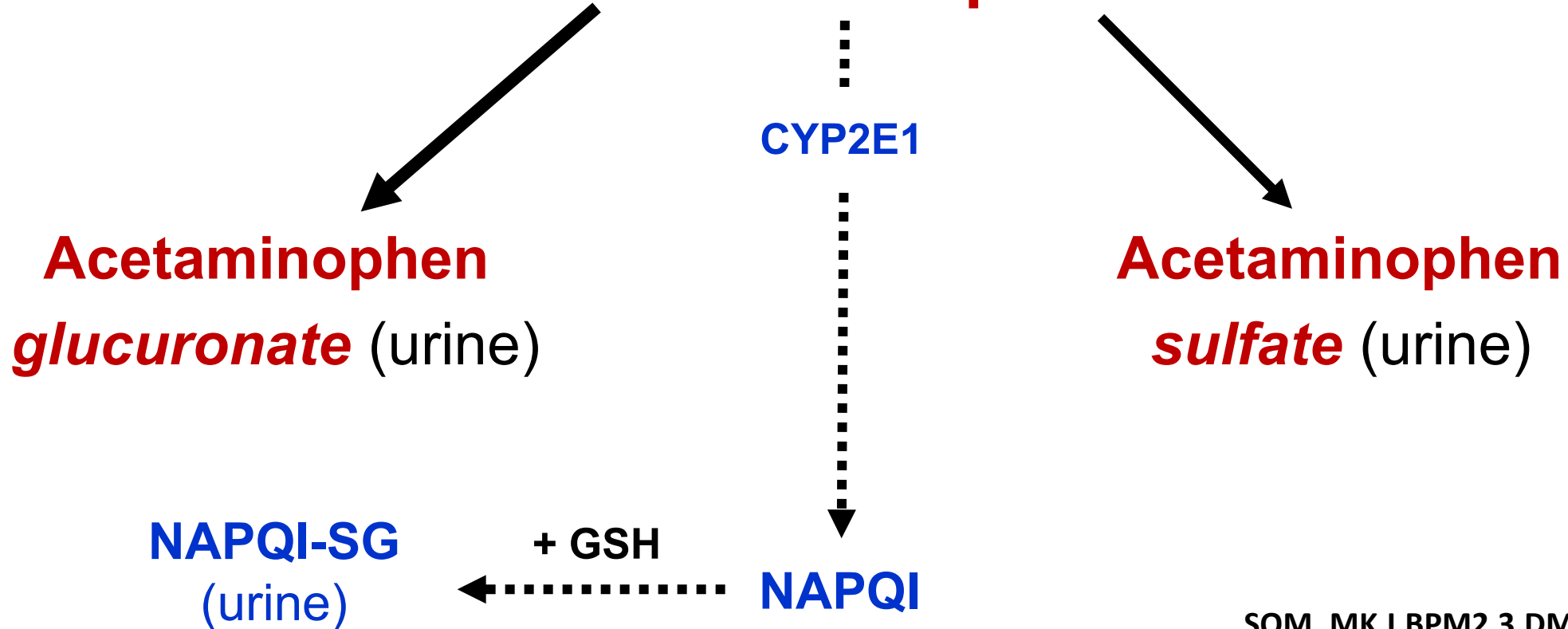
Grapefruit juice can change metabolism of many drugs (Statins) and can alter blood levels

Degradation of Acetaminophen

Conjugated with glucuronate or sulfate for excretion in urine

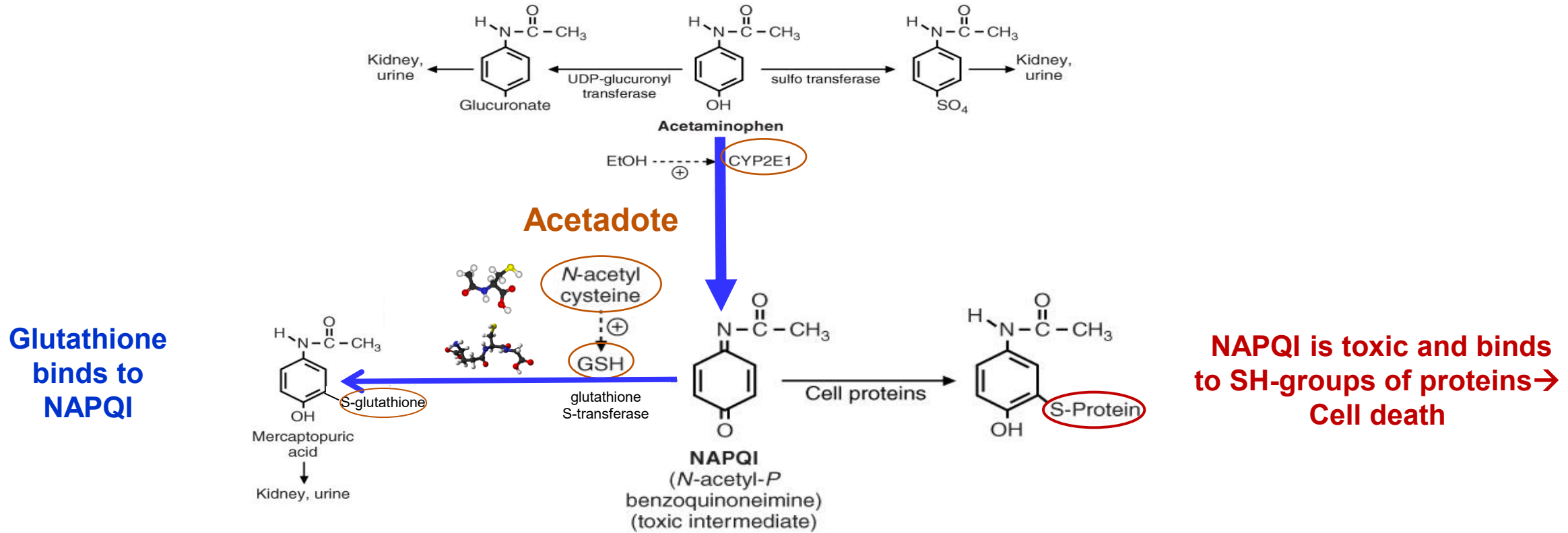


Acetaminophen



Toxicity of Acetaminophen: High NAPQI levels

Ethanol stimulates CYP2E1 synthesis



Acetadote (N-acetyl cysteine) prevents **severe** hepatic **damage** in **acetaminophen toxicity**



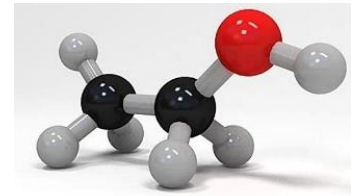
Ethanol



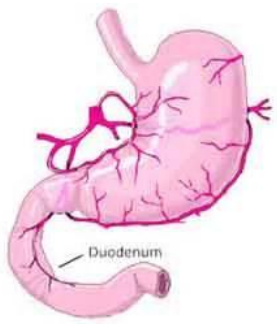
- **Excessive alcohol consumption is a public health problem.**

Alcoholic drinks contain different % of ethanol:

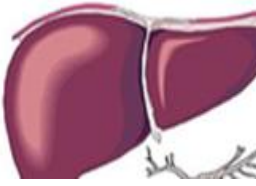
5% beer, **12%** wine and **40%** hard liquor by **volume**.



- Small molecule; **highly water** and **lipid soluble** and easily passes plasma membranes.
- Acts on **cerebral cortex**; **enhances** activity of **GABA receptors** and leads to **relaxation**.



Ethanol absorption and hepatic metabolism

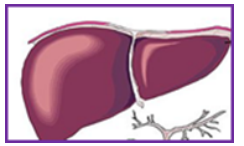


Ethanol is readily absorbed by passive diffusion

Uptake into blood is very efficient on an **empty stomach** (about **70%**) and in **duodenum**.

More than **90%** of **ethanol** enters blood and **metabolized** by **liver**
Metabolism differs at **low** or **high alcohol** concentration in **hepatocytes**.

Hepatocytes oxidize ethanol in **two steps**:
first to **acetaldehyde** and then to **acetate**



Ethanol
Alcohol DH ↓ ↓ **Microsomal ethanol oxidizing system**

Cytosol

Acetaldehyde

Mitochondria

Mitochondrial
acetaldehyde DH2

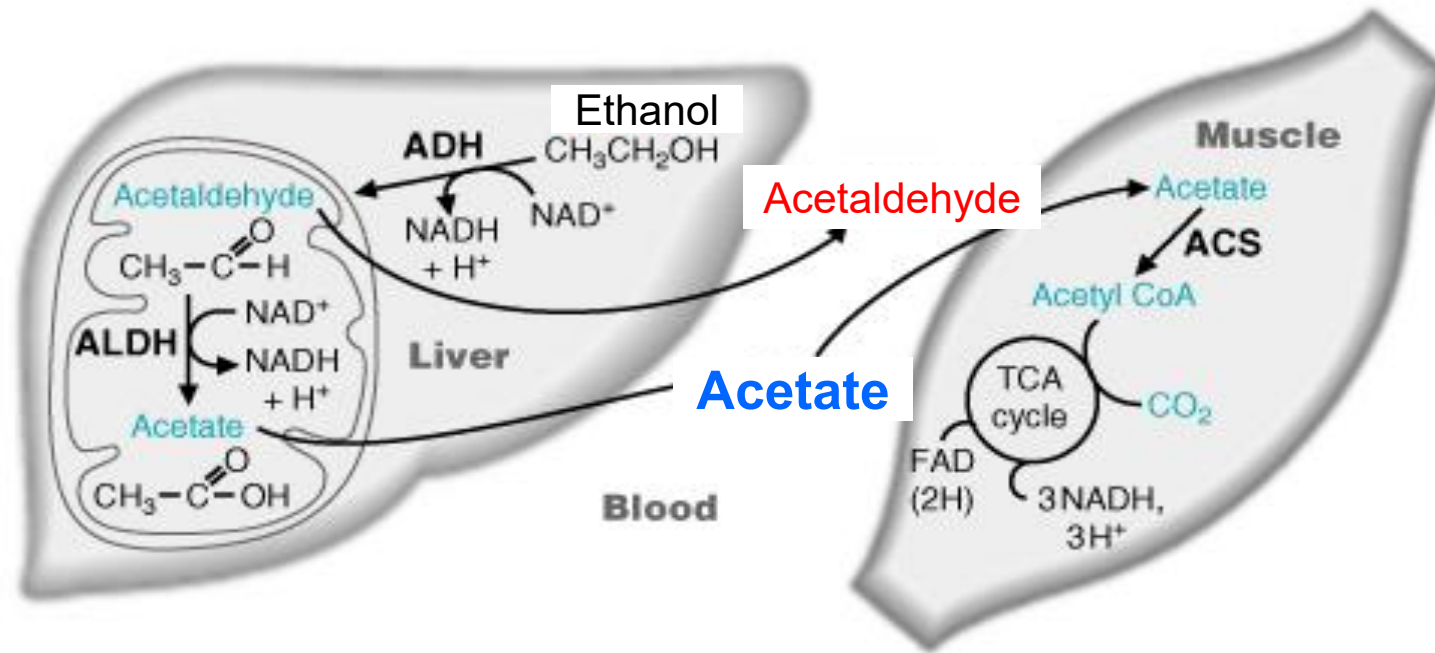
Cytosolic
acetaldehyde DH1

Acetate

Acetate is
released into the
blood

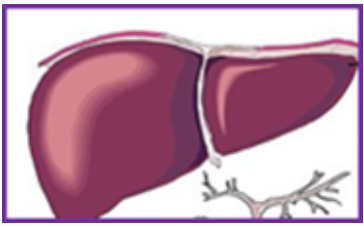


Fate of acetate



Acetate released into blood or can form ketone bodies in liver

Acetaldehyde is toxic to brain, heart, and other organs.



Low level of Ethanol

Alcohol dehydrogenase

Uses NAD^+ and forms NADH

Cytosol

ACETALDEHYDE

Mitochondria

Acetaldehyde DH2

Uses NAD^+ and forms NADH

Acetate

Acetate is
released
into the
blood



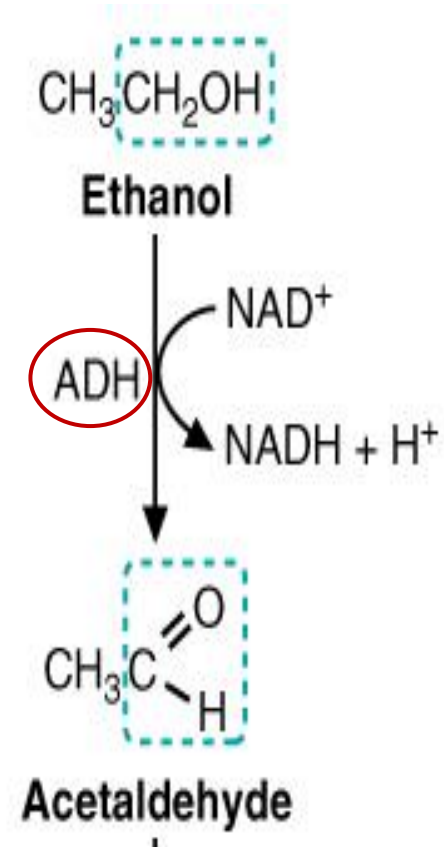
Alcohol dehydrogenase (ADH)

ADH at low alcohol levels in hepatocytes:

Ethanol is main substrate for alcohol dehydrogenase;
Forms NADH and acetaldehyde.

Alcohol dehydrogenase has high affinity for ethanol.
Isozymes with different **K_m** found in different individuals →
Hereditary difference in alcohol metabolism

Ethanol metabolism varies from individual to individual.



Marks p 459

Microsomal ethanol oxidizing system (MEOS) is active at high ethanol levels

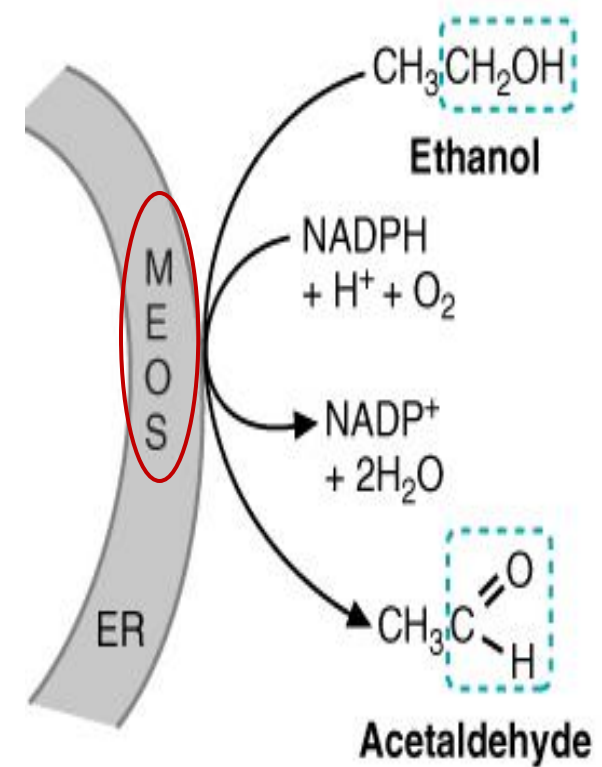
MEOS in hepatocytes:

Bound to **ER** membrane and forms **acetaldehyde** in **cytosol**.

MEOS has higher K_m (11 mM) and **lower affinity** for **ethanol** than alcohol dehydrogenase.

MEOS contains **CYP2E1** and **CYP3A4**.

Chronic alcohol consumption induces MEOS.



Marks p.459

Acetaldehyde is toxic

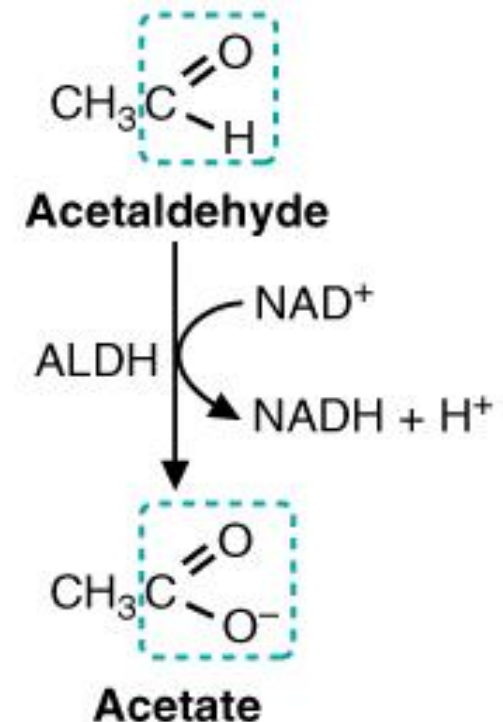
Metabolized to acetate (prevent damage)

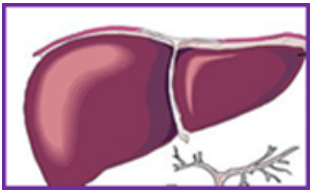
Acetaldehyde is substrate for **aldehyde dehydrogenases** (ALDH) and forms **acetate (non-toxic)** and **NADH**

Aldehyde DH isoforms in **mitochondria** and **cytosol**.

Mitochondrial aldehyde DH-2 has **high affinity** and catalyzes most of **acetaldehyde** oxidation.

Cytosolic aldehyde DH-1 has **low affinity** and **acts** when **cytosolic acetaldehyde** accumulates.





HIGH level of Ethanol in hepatocytes

Alcohol DH

High affinity
Forms NADH

MEOS

**Radical
formation**

Low affinity

Cytosol

↑ ACETALDEHYDE

Released
into blood

Acetaldehyde DH-1

Low affinity
Forms NADH

Mitochondria

Acetaldehyde DH-2

High affinity
Forms NADH

**Ketone body
synthesis**

Acetate

Acetate
released into
blood

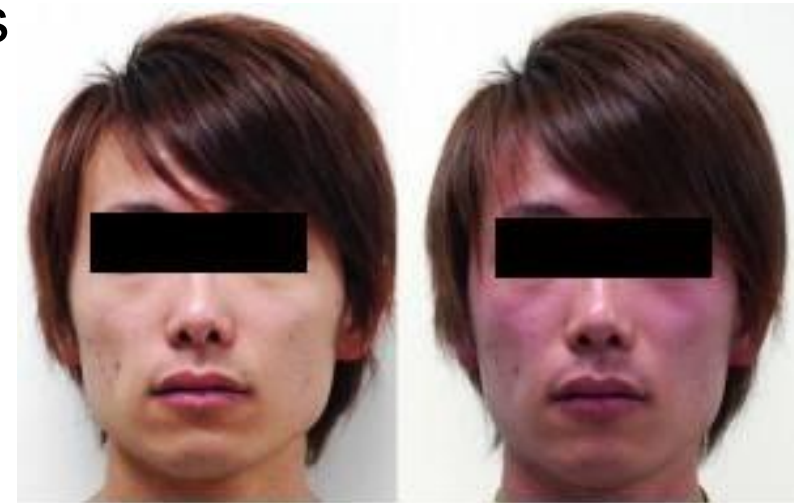


Hereditary deficiency of mitochondrial ALDH-2

An **allelic variant *ALDH2*2* (mitochondrial)** in individuals of **East Asian heritage** leads to decreased degradation of acetaldehyde in mitochondria → high acetaldehyde levels even at low ethanol intake.

High Acetaldehyde levels leads to:

- **Flushing** and **elevated heart rate**.
- **Nausea** and **vomiting**.

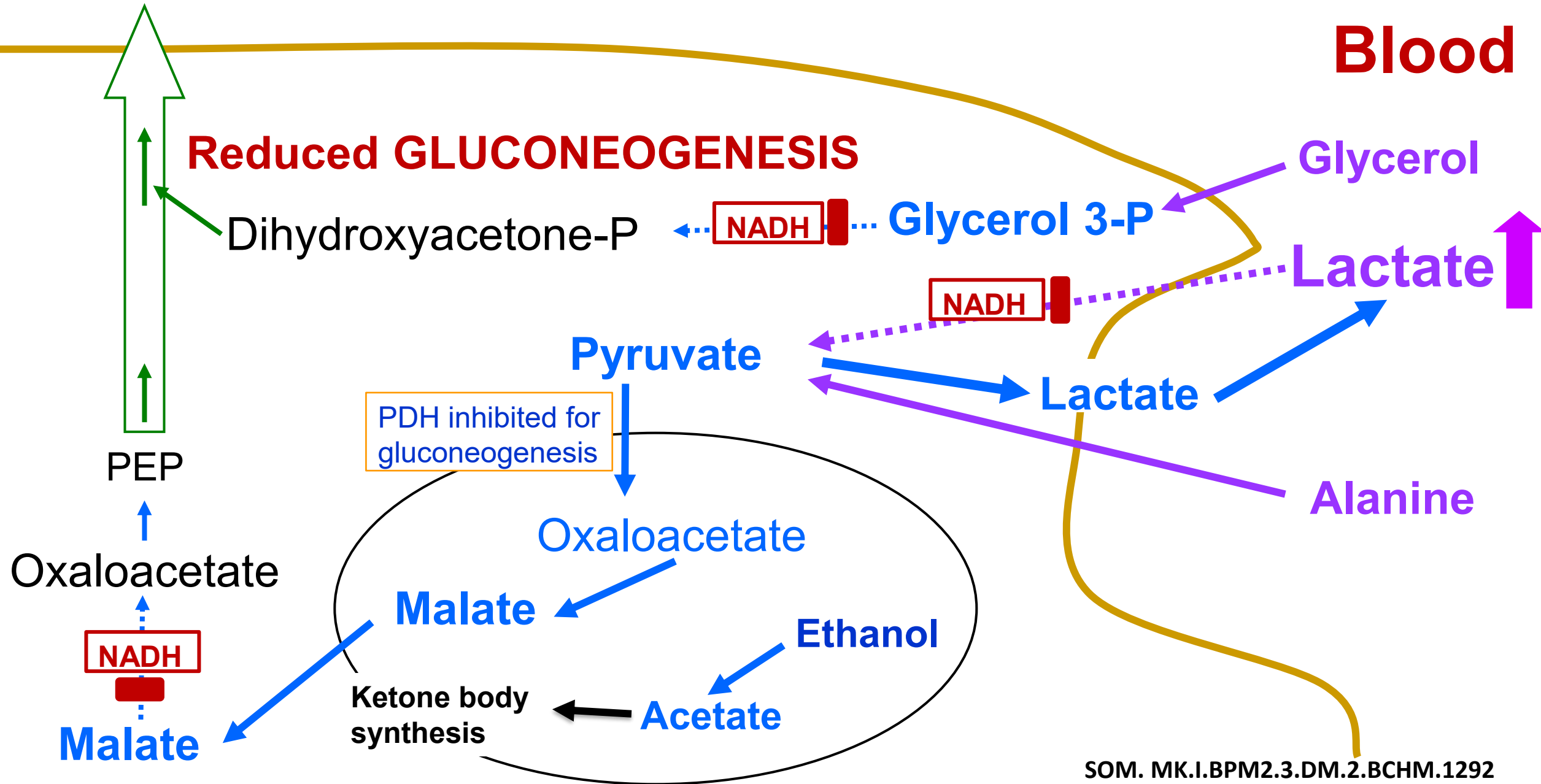


Asian Flush Syndrome

Complications of Alcohol Use

Acute	Chronic
Metabolic acidosis (ketones)	Alcohol-induced hypoglycemia
CNS depression, respiratory depression	Fatty liver disease, liver cirrhosis
Acetaldehyde toxicity	Wernicke-Korsakoff encephalopathy (confusion, ataxia, ophthalmoplegia) - Thiamine
Alcohol withdrawal (seizures)	Chronic pancreatitis
Acute pancreatitis	Vitamin deficiency (folate, B12)
Acute gastritis	Cardiomyopathy
Alcohol-induced hypoglycemia	Fetal alcohol syndrome

High ethanol intake: hypoglycemia, lactic acidemia and ketoacidosis



Text for previous slide:

High cytosolic NADH/NAD⁺ ratio reduces hepatic gluconeogenesis:
lack of NAD⁺ due to excessive ethanol degradation by ADH and ALDH.
Substrates for gluconeogenesis cannot be used

Glycerol cannot be used to form **DHAP** in cytosol.

Lactate cannot be used to form **pyruvate** at high NADH → Lactic acidemia

Alanine is **transaminated** to **pyruvate** → **Lactate** and released into **blood**.

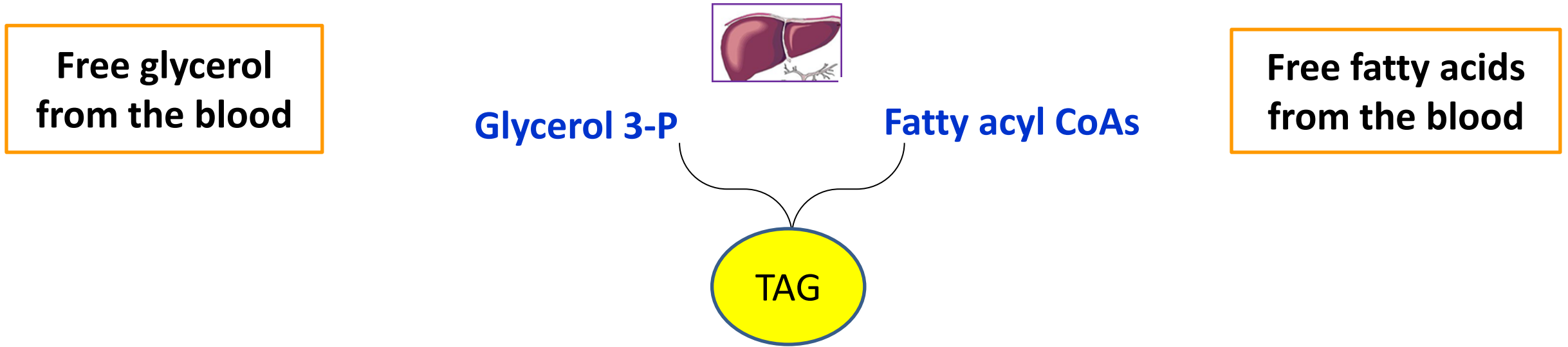
Oxaloacetate forms malate that goes into cytosol.

Malate cannot be used to form **cytosolic oxaloacetate**. (malate trapped)

PEP carboxykinase (gluconeogenesis) has less substrate.

Acetate formed is used for **ketogenesis (ketoacidosis)**

High levels of cytosolic NADH lead to TAG synthesis



- **Hepatic steatosis** (fatty change/ fatty liver) early and common consequence of chronic alcohol intake.
- Results from **acetaldehyde toxicity (inhibits tubulins)**
- Release of VLDL is inhibited and **lipid droplets (TAG)** accumulate inside **hepatocytes**.

Fatty liver: Toxicity of acetaldehyde

Tubulins needed for VLDL release into blood. **Tubulins damaged** by **acetaldehyde** leading to **lipid (TAG) accumulation** in **hepatocytes**.

ROS injury and lipid peroxidation

Binds to glutathione

ACETALDEHYDE

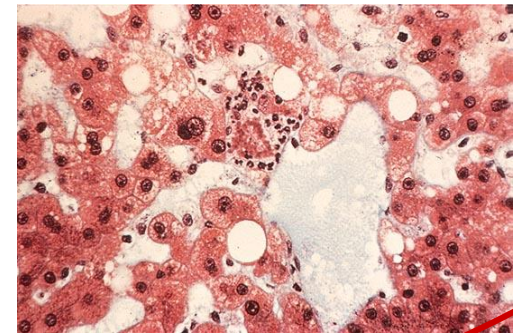
Inhibits release of VLDL
and serum proteins.

Liver: hepatic steatosis

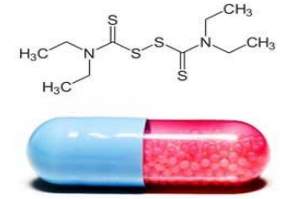
Blood: lack of blood clotting factors.

Alcoholic hepatitis:

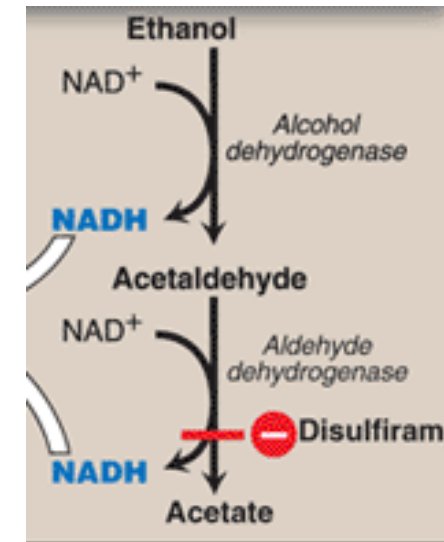
Liver biopsy shows marked **fatty infiltration** causing hepatic degeneration (ballooning of cells).



Disulfiram (Antabuse)



- Disulfiram (Antabuse) inhibits aldehyde dehydrogenase.
- Ingestion of **ethanol** leads to higher **acetaldehyde** levels when a patient takes **disulfiram**.
- **Alcohol** (small quantity) leads to: **flushing, nausea** (“hangover” symptoms)
- Intake of large amounts of **ethanol** can be very **dangerous**

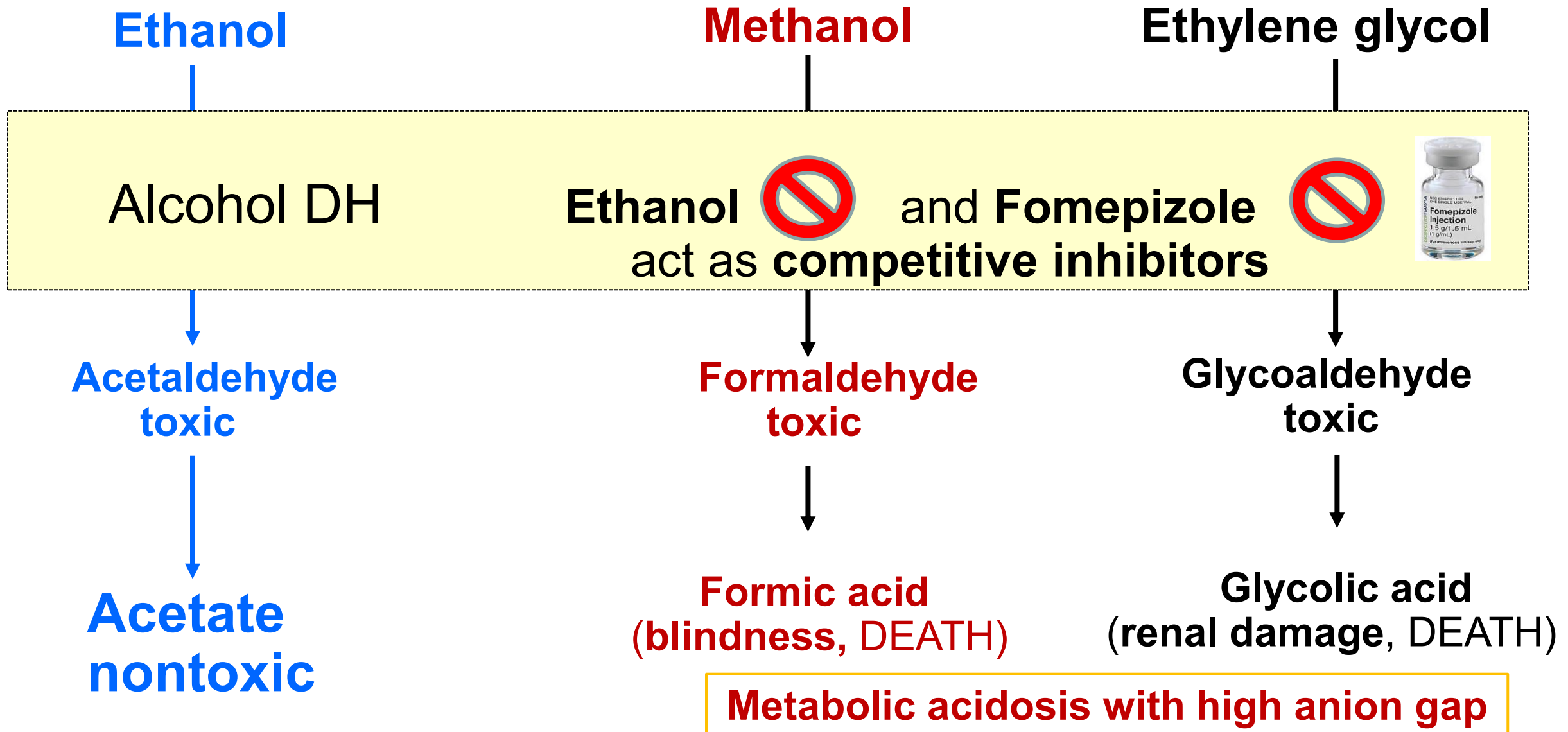


L.p.353.



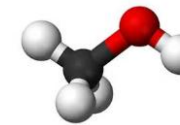
Metabolism of Methanol and ethylene glycol

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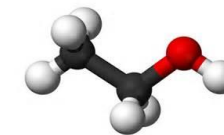




Methanol toxicity



Methanol



Ethanol

Methanol contaminant in alcoholic drinks

Oxidized by **alcohol dehydrogenase** to toxic **formaldehyde** in liver, neurons and **retina**

Formaldehyde and **formic acid** → CNS symptoms and **blindness** (“snowstorm vision”), **high anion gap metabolic acidosis**

Untreated methanol poisoning can lead to blindness and death.



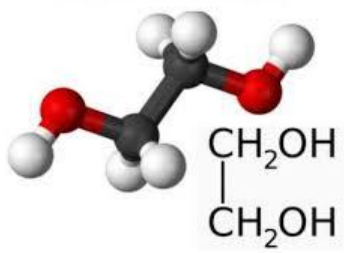
Treatment:

Alcohol dehydrogenase has higher affinity for ethanol than for methanol

Competitive inhibition of ADH by **ethanol** or **fomepizole**.

Metabolic acidosis treated with bicarbonate

Many patients need hemodialysis

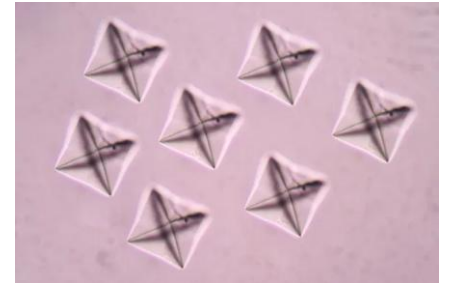


Ethylene glycol toxicity



Ethylene glycol in antifreeze (sweet taste)
Often ingested by children, cats, and dogs.

Ethylene glycol **oxidized** by **alcohol dehydrogenase** to highly toxic **glycoaldehyde** which forms **glycolic acid, oxalic acid** → **High anion gap metabolic acidosis**
Later **toxic calcium oxalate** causes **Renal failure** and death.



**Calcium oxalate
crystals in urine**

Treatment:

1. Inhibition of alcohol dehydrogenase by **fomepizole or ethanol**.
2. Treatment of metabolic acidosis
3. Hemodialysis if needed.

Methanol or ethylene glycol poisoning: High anion gap metabolic acidosis

- Anion gap used to classify metabolic acidosis.
- **Anion gap = Cations – Anions = $[\text{Na}^+] - ([\text{Cl}^-] + [\text{HCO}_3^-])$**

Normal anion gap range from 8 - 12 mEq/L

High anion gap Metabolic acidosis = pH less than 7.4; Low bicarbonate; Low PCO₂ (Anion gap is more than 20 mEq/L)

Metabolic acidosis: Low levels of bicarbonate.

HCO₃⁻ used to buffer anions (formate/ glycolate) → Increases Anion gap.

Poisoning with **methanol** or **ethylene glycol** leads to excess of **formic acid** or **glycolic acid** which are **nonvolatile acids/unmeasured anions**

Summary: Ethanol metabolism in hepatocytes

Ethanol forms **acetaldehyde**: **2 enzymes**

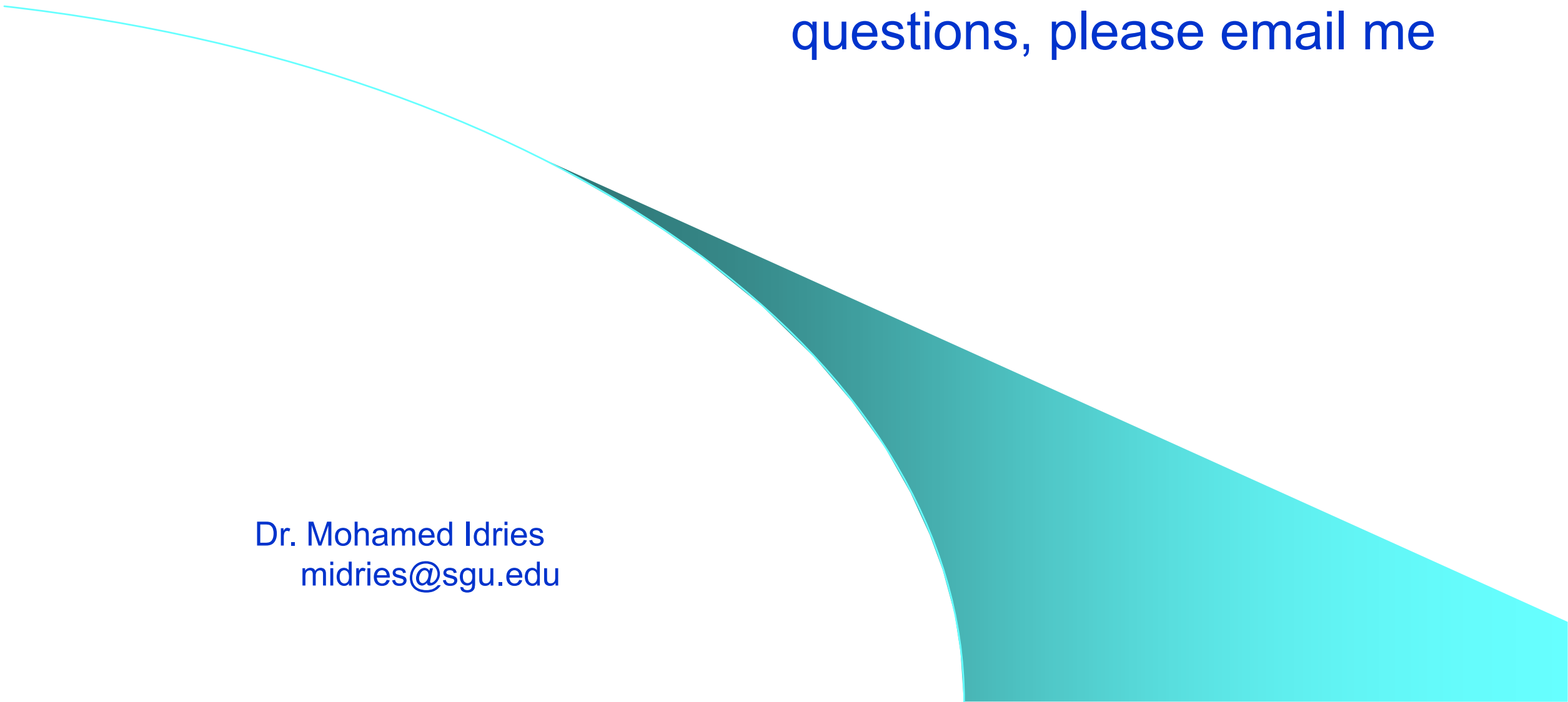
- **Alcohol dehydrogenase** (cytosol) forms **NADH**.
- **MEOS** (ER) generates **radicals**.

Acetaldehyde forms **acetate**: **2 enzymes**; Both form NADH.

- **Mitochondrial acetaldehyde dehydrogenase-2** (high affinity).
- **Cytosolic acetaldehyde dehydrogenase-1** (low affinity).

Hepatic ethanol metabolism leads to very high NADH/NAD⁺ ratio

Binge alcohol intake can result in hypoglycemia, ketoacidosis, and lactic acidosis

A decorative graphic element consisting of a curved shape that starts as a thin teal line on the left and expands into a larger, solid cyan area towards the bottom right corner of the slide.

Thank you if you have
questions, please email me

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